

Zoltrim 480 Oral Suspension

FOR ANIMAL USE ONLY

Product Name :

Zoltrim 480 Oral Suspension

Name and Strength of Active Substance(s)

Each ml contains 400mg Sulfamethoxazole, 80mg Trimethoprim and 10 mg benzyl alcohol (as preservative).

Product Description

Zoltrim 480 Oral Suspension is a white milky suspension upon agitation.

Pharmacodynamics

Alone, sulfonamides are bacteriostatic agents and trimethoprim is bactericidal, but in combination, the potentiated sulfas are bactericidal. Potentiated sulfas sequentially inhibit enzymes in the folic acid pathway, thereby inhibiting bacterial thymidine synthesis. The sulfonamide blocks the conversion of para-aminobenzoic acid (PABA) to dihydrofolic acid (DFA), and trimethoprim blocks the conversion of DFA to tetrahydrofolic acid by inhibiting dihydrofolate reductase. The potentiated Trimethoprim-Sulfonamides combinations have a relatively broad spectrum of activity. Gram-positive bacteria that are generally susceptible include most streptococci, many strains of staphylococci and Nocardia. Many gram-negative organisms of the family Enterobacteriaceae are also susceptible, but not *Pseudomonas aeruginosa*. Some protozoa (*Pneumocystis carinii*, Coccidia and Toxoplasma) are also inhibited by the combination. Resistance will develop slower to the combination of drugs than to either one alone.

Pharmacokinetics

Trimethoprim/sulfa is well absorbed following oral administration, with peak levels occurring about 1-4 hours after dosing. Trimethoprim/sulfa is well distributed in the body. When meninges are inflamed, the drugs enter the CSF in levels of about 50% of those found in the serum. Both drugs cross the placenta and are distributed into milk. Trimethoprim/sulfa is both renally excreted unchanged via glomerular filtration and tubular secretion and metabolized by the liver. The sulfas are primarily acetylated and conjugated with glucuronic acid and trimethoprim is metabolized to oxide and hydroxylated metabolites.

Indication

Pigs: Treatment and prevention of respiratory infections caused by *Actinobacillus pleuropneumoniae* susceptible to trimethoprim and sulfamethoxazole where the disease has been diagnosed in the herd.

Broilers: Treatment and prevention of respiratory infections caused by *Escherichia coli* susceptible to trimethoprim and sulfamethoxazole where the disease has been diagnosed in the flock.

Recommended Dosage

Pigs: 1 ml per 19.2 kg bodyweight per day or 1 litre in 2000 L drinking water for 3-4 days. This corresponds to 25 mg TMPS per kg bodyweight per day.

Broilers: 1 ml per 14.54 kg bodyweight per day or 1 litre in 3000 L drinking water for 3-4 days. This corresponds to 33 mg TMPS per kg bodyweight per day.

Mode of Administration

Shake well before use. To be administered via animal drinking water. Mixing Directions : ZOLTRIM 480 Oral Suspension should be added to clean drinking water to provide the desired final concentration. The liquid concentrate must be added in the amount of animal drinking water estimated to be consumed within 12 – 24 hours. Provide medicated water as the sole source of water each day until consumed, followed by non-medicated water as required. Fresh medicated water must be prepared each day.

Contraindications

Do not use in animals suffering from severe liver or kidney disease, oliguria or anuria.

Do not use in animals with impaired haematopoietic systems.

Do not use in case of known hypersensitivity to sulfonamides or trimethoprim.

Warnings and Precautions

Special warnings for each target species

Severely diseased animals can have a decreased appetite and water consumption. If necessary the product concentration in the drinking water should be adjusted to make sure that the recommended dosage is being consumed. However if the concentration of the product is increased too much, the intake of the medicated drinking water decreases for palatability reasons. Therefore water intake should be monitored regularly, especially in broilers. In case of insufficient uptake of water, pigs should be treated parenterally.

Special precautions for use in animals

Due to the likely variability (time, geographical) in susceptibility of bacteria for potentiated sulphonamides, occurrence of resistance of bacteria may differ from country to country and even from farm to farm, and therefore bacteriological sampling and susceptibility testing are recommended. Use of the product should be based on culture and sensitivity of micro-organisms from diseased cases on farm or from recent previous experience on the farm. Use of the product deviating from the instructions given in the SPC may increase the prevalence of bacteria resistant to sulfamethoxazole and trimethoprim and may also decrease the effectiveness of combinations of trimethoprim with other sulphonamides due to the potential for cross resistance.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

During preparation and administration of the medicated drinking water, skin contact with the drug should be avoided. Therefore it is recommended to wear impermeable gloves e.g. rubber or latex when applying the product. Do not handle this product if you know you are allergic to trimethoprim or sulphonamides. Do not smoke, drink or eat when handling the product. In the event of eye contact, rinse the eye with copious amounts of clean water and if irritation occurs, seek medical attention. In the event of accidental ingestion, seek medical advice. Wash hands and contaminated skin immediately after handling the product. If you develop symptoms following exposure to the product such as skin rash, you should seek medical advice and show the present warning to the doctor. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

Interactions with Other Medicaments

Do not combine with other veterinary medicinal products.

Safe Use During Pregnancy/Lactation

Safety of trimethoprim/sulfa has not been clearly established in pregnant animals. Use during pregnancy only if risks outweigh the benefits since folic acid metabolism may be affected. Sulfonamides are excreted in low concentrations in breast milk. Safety of trimethoprim/sulfa has not been clearly established in lactating animals.

Side Effects

A diminished water intake in chickens may occur occasionally. Hypersensitivity reactions can occur rarely.

Overdose and Treatment

A 2 ½ fold overdose is well tolerated by pigs. In chicken an acute overdose will not occur because the birds will be reluctant to drink the strongly concentrated drinking water (too bitter taste if above 2 litres of the veterinary medicinal product per 1000 litre drinking water). Chronic overdose in chicken will result in a strongly diminished water- and feed intake and retarded growth.

Withdrawal Period

Meat and offal : Pigs : 5 days ; Broilers: 6 days

Eggs: Not authorised for use in laying birds producing eggs for human consumption.

Shelf-life

As packaged for sale : 3 years.

After first opening of the immediate packaging : 24 hours.

After dilution or reconstitution according to directions : 24 hours.

Allowable Maximum Residual Limit (MRL)

Sulfamethoxazole:(Pig) Muscle, fat, liver, kidney and edible tissue=20µg/kg. (Chicken) Muscle, liver, kidney and edible tissue = 20µg/kg, fat = 50µg/kg, [Ref : The Japan Food Chemical Research Foundation]

Trimethoprim:(Pig) Muscle, skin + fat, liver, kidney = 50µg/kg. (Poultry) Muscle, skin + fat, liver, kidney = 50µg/kg, [Ref : EMEA/MRL/828/02-FINAL, January 2002]

Storage Conditions

Do not store above 25°C. Keep out of reach of children. Jauhkan dari kanak-kanak.

Disposal of containers

Empty bottle must be safely disposed in trash according to local government regulations.

Packaging

Available in 1 litre per bottle, pack in 12 bottles or 20 bottles per carton.

Name and Address of manufacturer

Life Biopharma Sdn. Bhd. (546163-D)

No. 250-251, Jalan Industri Galla 13, Kawasan Perindustrian Galla, 70200 Seremban, Negeri Sembilan, Malaysia.

Date of Revision of Package Insert

2014-10-17